

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0895

Drewry 100 kV
December 17, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0895. The Interconnection Customer proposes a 121.5 MW solar photovoltaic facility interconnecting at the Drewry 100 kV substation in the Indiana area. The request was received on November 21, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 252 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,868,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CUBA CITY – FARMERSVILLE 100 kV
Contingency: DEKALB – JONES CHAPEL 500 kV
Loading: 120.9% (Rating: 322 MVA)
DFAX: 6.5%
Pre-project loading: 118.5%
2. Facility: ROBISON ARK – RUSSELL FORK 230 kV
Contingency: JONES CHAPEL – DEKALB 500 kV
Loading: 120.7% (Rating: 2,461 MVA)
DFAX: 45.9%
Pre-project loading: 118.4%
3. Facility: GOLDEN GATE – LAOTTO 100 kV
Contingency: JONES CHAPEL – DEKALB 500 kV L
Loading: 118.9% (Rating: 282 MVA)
DFAX: 6.5%
Pre-project loading: 116.1%
4. Facility: KENDALLVILLE – REX BROWN 100 kV
Contingency: DEKLB – JONES CHAPEL 500 kV Line
Loading: 117.3% (Rating: 361 MVA)
DFAX: 6.5%
Pre-project loading: 115.1%
5. Facility: PUEBLO XTRA 500/230kV XFMR
Contingency: PUEBLO XTRA – BOCA RATON 500 kV
Loading: 115.9% (Rating: 1,916 MVA)
DFAX: 21.2%
Pre-project loading: 114.5%

6. Facility: REX BROWN – MONTIBELLO 100 kV
Contingency: DEKALB – JONES CHAPEL 500 kV
Loading: 115.7% (Rating: 380 MVA)
DFAX: 6.5%
Pre-project loading: 113.7%
7. Facility: LIGONIER – MONTIBELLO 100 kV
Contingency: JONES CHAPEL – DEKALB 500 kV
Loading: 114.7% (Rating: 394 MVA)
DFAX: 6.5%
Pre-project loading: 112.7%
8. Facility: RICHLAND – DEKALB 500 kV
Contingency: DEKALB – SLOCUM RD 230 kV line
Loading: 114.5% (Rating: 3,591 MVA)
DFAX: 51.3%
Pre-project loading: 112.8%
9. Facility: LARSON STREET – CORRIDOR 230 kV L
Contingency: BEAVER – TREGO 500 kV Line
Loading: 111.6% (Rating: 851 MVA)
DFAX: 9.8%
Pre-project loading: 110.2%
10. Facility: SEGS 5 – HILLCREST 100 kV L
Contingency: RICHLAND – DEKALB SUBSTATION 500 kV line
Loading: 111.5% (Rating: 554 MVA)
DFAX: 7.3%
Pre-project loading: 109.9%
11. Facility: LOOKINGGLASS – TRUMAN 230 kV
Contingency: TREGO – BEAVER 500 kV
Loading: 110.8% (Rating: 911 MVA)
DFAX: 9.8%
Pre-project loading: 109.5%
12. Facility: BARAGA VLG – GRAY'S FLAT 230 kV
Contingency: EAST METHUEN – MADELINE 230 kV L
Loading: 109.8% (Rating: 1,020 MVA)
DFAX: 9.1%
Pre-project loading: 108.8%
13. Facility: TREGO SUBSTATION – HIGHGATE 230 kV Line
Contingency: TROY ENERGY LLC – TREGO 500 kV
Loading: 109.8% (Rating: 652 MVA)
DFAX: 10.2%
Pre-project loading: 107.8%
14. Facility: TRUMAN – WALDO 230 kV L
Contingency: BEAVER – TREGO 500 kV
Loading: 109.1% (Rating: 949 MVA)
DFAX: 9.8%
Pre-project loading: 107.8%
15. Facility: NAPOLEON PEAKING STATION – RICHLAND 230 kV
Contingency: RICHLAND – TROY ENERGY LLC 500 kV line
Loading: 108.8% (Rating: 1,063 MVA)
DFAX: 12.2%
Pre-project loading: 107.4%
16. Facility: TROY ENERGY LLC – RICHLAND 500 kV Line
Contingency: TREGO – SWAN VLY 500 kV L

- Loading: 108.2% (Rating: 3,510 MVA)
DFAX: 36.0%
Pre-project loading: 106.9%
17. Facility: GRAY'S FLAT – E METHUEN 230 kV
Contingency: MADELINE – EAST METHUEN 230 kV line
Loading: 108.0% (Rating: 1,052 MVA)
DFAX: 9.1%
Pre-project loading: 107.0%
18. Facility: JONES CHAPEL 500/230kV XFMR
Contingency: DEKALB – JONES CHAPEL 500 kV
Loading: 107.5% (Rating: 2,166 MVA)
DFAX: 14.3%
Pre-project loading: 106.7%
19. Facility: MADELINE – BARAGA VLG 230 kV Line
Contingency: GRAY'S FLAT – EAST METHUEN SUBSTATION 230 kV
Loading: 106.9% (Rating: 637 MVA)
DFAX: 6.6%
Pre-project loading: 105.6%
20. Facility: BURRIS PIT – HEBO 230 kV
Contingency: TREGO – BEAVER 500 kV
Loading: 105.7% (Rating: 1,021 MVA)
DFAX: 8.0%
Pre-project loading: 104.8%
21. Facility: LOOKINGGLASS – CORRIDOR 230 kV line
Contingency: TREGO – BEAVER 500 kV
Loading: 105.7% (Rating: 926 MVA)
DFAX: 9.8%
Pre-project loading: 104.4%
22. Facility: TREGO 230 kV/1 kV Transformer
Contingency: TREGO – TROY ENERGY LLC 500 kV line
Loading: 105.6% (Rating: 829 MVA)
DFAX: 12.4%
Pre-project loading: 103.8%
23. Facility: BURRIS PIT – JAMES DE YOUNG 230 kV
Contingency: BEAVER – TREGO 500 kV
Loading: 105.3% (Rating: 1,046 MVA)
DFAX: 8.0%
Pre-project loading: 104.3%
24. Facility: SABANERA 500-230 kV TX
Contingency: LONETREE – SABANERA 500 kV
Loading: 104.9% (Rating: 1,302 MVA)
DFAX: 5.3%
Pre-project loading: 104.4%
25. Facility: MINNEHAHA – STA S VAN NUYS 230 kV
Contingency: TROY ENERGY LLC – RICHLAND 500 kV
Loading: 104.7% (Rating: 1,336 MVA)
DFAX: 16.3%
Pre-project loading: 103.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CUBA CITY – FARMERSVILLE 100 kV	Rebuild line with higher-capacity conductor	\$7,951,000	1% (\$103,000)
ROBISON PARK – RUSSELL FORK 230 kV	Rebuild line with higher-capacity conductor	\$37,065,000	1% (\$427,000)
GOLDEN GATE – LAOTTO 100 kV L	Rebuild line with higher-capacity conductor	\$5,241,000	1% (\$68,000)
KENDALLVILLE SUBSTATION – REX BROWN 100 kV L	Rebuild line with higher-capacity conductor	\$2,710,000	1% (\$35,000)
PUEBLO XTRA 500/230kV XFMR	Replace transformer with higher-rated unit	\$54,222,000	1% (\$389,000)
REX BROWN – MONTIBELLO 100 kV	Rebuild line with higher-capacity conductor	\$5,938,000	1% (\$77,000)
MONTIBELLO – LIGONIER 100 kV Line	Rebuild line with higher-capacity conductor	\$5,791,000	1% (\$75,000)
RICHLAND – DEKALB 500 kV	Rebuild line with higher-capacity conductor	\$160,875,000	1% (\$1,671,000)
LARSON STREET – CORRIDOR 230 kV L	Rebuild line with higher-capacity conductor	\$11,584,000	1% (\$102,000)
SEGS 5 – HILLCREST 100 kV L	Rebuild line with higher-capacity conductor	\$3,270,000	1% (\$34,000)
LOOKINGGLASS – TRUMAN 230 kV	Rebuild line with higher-capacity conductor	\$19,806,000	1% (\$174,000)
BARAGA VLG – GRAY'S FLAT 230 kV	Reconductor line	\$12,658,000	1% (\$100,000)
TREGO SUBSTATION – HIGHGATE 230 kV Line	Reconductor line	\$10,484,000	1% (\$113,000)
TRUMAN – WALDO 230 kV L	Reconductor line	\$6,118,000	1% (\$54,000)
NAPOLEON PEAKING STA – RICHLAND 230 kV	Reconductor line	\$8,773,000	1% (\$91,000)
TROY ENERGY LLC – RICHLAND 500 kV Line	Reconductor line	\$112,166,000	1% (\$1,069,000)
GRAY'S FLAT – E METHUEN 230 kV	Reconductor line	\$13,961,000	1% (\$110,000)
JONES CHAPEL 500/230kV XFMR	Replace transformer with higher-rated unit	\$22,224,000	0% (\$113,000)
MADELINE – BARAGA VLG 230 kV Line	Reconductor line	\$6,792,000	1% (\$53,000)
BURRIS PIT – HEBO 230 kV	Reconductor line	\$7,189,000	1% (\$58,000)
LOOKINGGLASS – CORRIDOR 230 kV line	Reconductor line	\$12,348,000	1% (\$108,000)
TREGO 230 kV/1 kV Transformer	Replace transformer with higher-rated unit	\$10,964,000	1% (\$114,000)
BURRIS PIT SUBSTATION – JAMES DE YOUNG 230 kV	Reconductor line	\$20,250,000	1% (\$162,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SABANERA 500-230 kV TX	Replace transformer with higher-rated unit	\$56,248,000	1% (\$501,000)
MINNEHAHA – STATION S VAN NUYS 230 kV Line	Reconductor line	\$6,544,000	1% (\$67,000)
	Total allocated to project		\$5,868,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.